

ARUN ARUNACHALAM

Jersey City, NJ | 908-330-1050 | asnarun75@gmail.com | linkedin.com/in/arunarunachalam75

PROFESSIONAL SUMMARY

Senior Director-caliber AI and software engineering leader with 17+ years delivering large-scale platform transformation across capital markets and enterprise data systems at Bank of America. Deep hands-on expertise building production LLM-based applications, RAG pipelines, and agentic AI systems, combined with a proven track record of owning end-to-end modernization roadmaps and driving measurable business outcomes against firm-wide commitments. Brings extensive experience with distributed systems, event-driven and batch architectures, and cloud-native platforms (AWS, Kafka, Kubernetes) in 24/7 financial services environments. Equally effective writing code alongside engineering teams, presenting investment decisions to executive audiences, and coordinating cross-team governance across complex, multi-domain programs.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

GenAI & Agentic AI Systems (RAG Pipelines, Multi-Agent Orchestration, LLMOps) | AI-Assisted SDLC & Modernization Toolchains | Cloud-Native Architecture (AWS, EKS, Kubernetes) | Event-Driven & Batch Systems (Kafka) | Distributed Systems (24/7 High-Availability) | Platform Modernization at Scale | Data Engineering (Spark, Databricks, ETL/ELT) | Trading, Risk & Capital Markets Platforms | Executive Stakeholder Leadership | Global Engineering Team Leadership | Regulatory & Compliance Systems (FRB, FINRA, SEC)

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Designed and operated end-to-end AI modernization workflows, including artifact ingestion, ML-based anomaly detection, and AI-assisted specification and code generation pipelines — delivering **35% faster delivery** and **45% productivity gain** across engineering teams
- Built and deployed production GenAI platforms leveraging RAG pipelines and agentic AI systems to accelerate SDLC workflows and enable intelligent decision systems in 24/7 capital markets environments
- Established LLMOps practices for AI toolchain deployment, monitoring, and continuous improvement across mission-critical processing systems
- Owned strategic sequencing and roadmap for enterprise-scale platform transformation across trading and data systems supporting **\$1T+ daily processing**, establishing migration beachheads and scaling against business commitments
- Modernized legacy systems into cloud-native, event-driven architecture (AWS, Kafka, Kubernetes), improving throughput by **40%** while maintaining **99.97% uptime**

- Delivered real-time trading, pricing, and risk platforms across multiple domains, partnering directly with domain subject-matter experts to validate migrated logic and preserve business intent
- Coordinated cross-team governance across technology leads, architecture teams, and senior leadership — presenting program progress, trade-offs, and investment decisions to both technical and executive audiences
- Built and led global engineering teams driving **15% velocity improvement**, hiring, mentoring, and retaining senior technical talent across US, UK, and India
- Ensured regulatory compliance (FRB, FINRA, SEC) across mission-critical systems in 24/7 production environments

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Led consolidation of risk systems across Credit and structured products domains, contributing to **~\$2B capital efficiency** — directly analogous to multi-domain migration sequencing and business-intent preservation
- Built real-time pricing and analytics platforms for structured products and mortgage trading, collaborating with architecture teams to define target-state platforms
- Managed cross-functional program delivery across business and technology stakeholders, establishing engineering standards and governance frameworks

Bank of America — AVP, Whole Loan Trading Desk Support (*prior to 2011*)

- Architected and delivered end-to-end Java EE trading support applications, designing API and distributed system components supporting high-volume batch and event-driven processing
- Introduced CI/CD practices reducing release cycle from monthly to weekly, establishing foundational DevOps and engineering standards adopted across the organization

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building production GenAI platforms leveraging RAG pipelines and agentic AI systems for real-world, 24/7 financial services applications
- Designing multi-agent orchestration systems for discovery, specification generation, and automated verification — directly applicable to legacy code ingestion and migration acceleration
- Establishing LLMOps practices including deployment, monitoring, cost management, drift detection, and continuous improvement for AI toolchains
- Applied ML for anomaly detection and operational intelligence across distributed, high-availability platforms
- Driving AI-assisted SDLC workflows that measurably improve delivery speed, code quality, and engineering productivity

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Leadership: Executive stakeholder management, global engineering team leadership, cross-team governance, program roadmap ownership, P&L ownership, strategic planning, Agile, SDLC, ITIL, SOX, FRB/FINRA/SEC compliance

AI & GenAI: RAG pipelines, agentic AI systems, multi-agent orchestration, LLMOps (deployment, monitoring, drift detection, cost management), ML anomaly detection, AI-assisted SDLC, production LLM-based applications

Architecture: Cloud-native architecture (AWS, EKS, Kubernetes), event-driven systems (Kafka), batch architectures, distributed systems (24/7 high-availability), enterprise architecture, API design (REST/SOA)

Data Engineering: Spark, Databricks, ETL/ELT, Snowflake, Hadoop, Cassandra, MongoDB

Engineering: Java, Python, Node.js, SQL, Oracle, Sybase, WebLogic, Kafka, Kubernetes, AWS

Tools: ServiceNow, JIRA, Git, Visio, Microsoft Project